

The What if? collaborative planning support system

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Abstract. In this paper I will describe What if?, a scenario-based, policy-oriented planning support system (PSS) that uses increasingly available GIS data to support community-based processes of collaborative planning and collective decisionmaking. It incorporates procedures for conducting land suitability analysis, projecting future land-use demands, and allocating the projected demands to the most suitable locations. The system allows users easily to create alternative development scenarios and determine the likely impacts of alternative public-policy choices on future land-use patterns and associated population and employment trends. It is easy to use, can be customized to the user's database and policy issues, and provides outputs in easy to understand maps and reports.

Introduction

The field of urban and regional planning is experiencing two fundamental changes that are having a profound impact on the use of computer-based models in planning practice and education. On the one hand, the dramatically increased availability of powerful, low-cost, and easy to use GIS tools and more extensive spatially referenced data are making GIS an essential tool for planning tasks such as land-use monitoring, code enforcement, and permit tracking. However, with this increased use has come an increased realization that GIS alone cannot serve all of the needs of planning (for example, see Couclelis, 1991; Harris, 1989; Harris and Batty, 1993). This realization has renewed planners' interest in computer modeling and stimulated the development of planning support systems (PSS) which combine GIS (and non-GIS) data, computer-based models, and advanced visualization techniques into integrated systems to support core planning functions such as plan preparation and evaluation (for example, see Finaly and Marples, 1992; Holmberg, 1994; Klosterman, 1997).

On the other hand, planners are increasingly abandoning the attempt to 'plan for' the public and opting for more collaborative efforts to 'plan with' the public. These attempts take a variety of forms such as strategic visioning conferences, design charrettes, and other community-based collaborative planning and consensus-building efforts (for example, see Goldschalk et al, 1994; Lowry et al, 1997). These efforts attempt to involve private citizens and other stakeholders directly in the planning process to help identify and prioritize the public's needs and desires, explore alternative development scenarios, and establish benchmarks for evaluating ongoing development efforts. As an ideal, these carefully structured consensus-building processes can help achieve the ideals of communicative rationality and traditional comprehensive land-use planning (Healey, 1992; Innes, 1992; 1996).

What if? is an interactive GIS-based PSS that responds directly to both of these trends. It uses the GIS data sets that communities have already developed to support community-based efforts to evaluate the likely implications of alternative public-policy choices. It provides a single integrated package that allows planning tasks that at present require weeks or even months to do to be completed quickly and easily.

The package can be customized to a community's existing GIS data, concerns, and desires, and provides outputs in easy to understand maps and reports which can be used to support community-based collaborative planning efforts.⁽¹⁾

Reflecting the movement from 'planning for the public' to 'planning with the public', What if? assumes that planning and policymaking should not be based on closed and unsupervised 'objective' analysis of technical experts. Instead it provides a set of computer-based tools which facilitate open and ongoing processes of community learning, debate, and compromise. By providing information, techniques, and evaluation criteria which can be used to support particular public-policy perspectives and critique others, it supports explicitly political processes of 'adversary' or 'counter' modeling similar to Davidoff's (1965) advocacy planning model. By involving the public directly in the planning process, it helps constrain the discretion of professionals and political insiders, reduces the knowledge differential between professionals and laypersons, and gives the public a sense of ownership in the plans and proposals which are developed.

Acknowledging the limitations of planners' knowledge, information, and resources, the model abandons the unrealistic goal of producing a single 'exact' prediction of the future for the preparation of a range of alternative scenario-based forecasts which reveal a range of potential futures. With the recognition that models are useful in a policy context only if they are understood by policymakers, the underlying structure, relationships, and assumptions of What if? are made as explicit and clear as possible. Most importantly, as its name suggests, the model adopts the 'what if' metaphor popularized by electronic spreadsheets by acknowledging that the model results identify only *what* would happen *if* the underlying assumptions in a scenario are correct. It does this by explicitly identifying alternative policy choices, allowing users to choose between these alternatives and to determine their likely effects on the area being studied. Similarly, the underlying assumptions of the model concerning the 'state of nature' are explicitly stated and easily modified, allowing users to determine the effect that these assumptions have on the model results (Klosterman, 1987, pages 444–448).

The remainder of the paper is divided into four sections. The next section provides an overview of the modeling process and the application areas for which What if? is most appropriate. The following section describes the three analysis steps that are incorporated in the model, the procedures for conducting each step, and the associated model outputs. The second to last section identifies the information and procedures that are required to adapt the system for use in a particular study area. The paper concludes with an evaluation of the strengths and weaknesses of the model.

System overview

As its name suggests, What if? does not attempt to predict future conditions exactly. Instead it is an explicitly policy-oriented planning tool that can be used to determine *what* would happen *if* clearly defined policy choices are made and assumptions concerning the future prove to be correct. Policy choices that can be considered in the model include the staged expansion of public infrastructure and the implementation of alternative land-use plans or zoning ordinances. Assumptions for the future that can be considered in the model include future population and employment trends, assumed household characteristics, and anticipated development densities.

⁽¹⁾ What if? was developed by the Plan it!, LLC, a cooperative venture of Community Analysis and Planning Systems Inc., Data Chromatics Inc., and LDR International. The first version of the software was developed to support the preparation of the Western Hamilton County Comprehensive Plan for a rapidly growing area to the west of Cincinnati, Ohio. The model is fully operational and documented. A demonstration version is also available. Please contact the author for information on the current status of the model.

What if? is a 'bottom-up' model which begins with homogeneous land units or uniform analysis zones (UAZs), applies alternative policy choices to these units, allocates projected land-use demands to them, and then derives regional conditions (such as population and employment growth trends) by aggregating the values for these land units.⁽²⁾ UAZs are GIS-generated polygons which are homogeneous in all respects considered in the model. Thus, for instance, all points within a UAZ have the same slope, are located in the same municipality, have the same zoning designation, are within the same distance of an existing or proposed highway, and so on.

UAZs are created for a study area by using GIS overlay functions to combine all of the relevant layers of information on natural and human-made features. The map layers can contain information on natural conditions (for example, slopes, soils, and scenic vistas), existing and proposed infrastructure (for example, the proximity to intersections or major roads and the availability of sewer and water service), and land-use controls such as zoning districts and planned land uses. The UAZs contain information which was provided in each of the constituent layers, that is, each UAZ contains information on the slope, the availability of sewer and water service, planned land use, and so on, for all points lying inside of it.

As will be described in detail below, What if? projects future land-use patterns by balancing the supply of, and demand for, land suitable for different uses at different locations. Alternative visions for an area's future can be explored by defining alternative suitability, growth, and allocation scenarios. For example, a 'Trend' scenario could determine the effects of continuing current development policies. An 'Environmental Protection' scenario might consider the impact of policies that severely limit growth in scenic areas and on land highly suitable for agricultural uses. A 'Build out' scenario would reveal the implications of allowing growth to continue until it reached permitted density levels for all developable parcels in the study area. The assumptions underlying these and other scenarios can easily be modified to incorporate the full range of alternative visions for an area's future.

The results generated by considering these alternative scenarios provide concrete and understandable expressions of the likely results underlying the policy choices and assumptions of a scenario. For instance, the model might demonstrate that there is insufficient land simultaneously to accommodate high growth, low residential densities, and strict agricultural protection policies, forcing the community to choose between highly desirable, but inconsistent, policy goals.

What if? is most appropriate for areas that are experiencing, or anticipating, rapid urbanization and the associated problems of traffic congestion, inadequate public infrastructure, and the loss of agricultural and open land. Areas that are currently undeveloped and will remain so in the future have few impacts and policy options to consider. Currently developed areas face complex issues of redevelopment and reuse that are extremely difficult to capture in a computer model. Areas on the urban fringe experiencing rapid and uncontrolled urbanization and industrialization face difficult and complex issues of managing growth and maintaining a sound fiscal base. As a result, What if? is particularly useful in promoting public dialogue and collective decisionmaking in these rapidly changing areas.

What if? was developed with Microsoft's Visual Basic and the Environmental Sciences Research Institute's MapObjects GIS component software. It incorporates many of the design concepts in the first California Urban Futures (CUF) model (Landis, 1994; 1995) and similar models such as the Sophisticated Allocation Process (SOAP) model of the

⁽²⁾ The term 'uniform analysis zone' (UAZ) is used instead of Landis's (1994) comparable concept of 'developable land unit' (DLU) because these spatial units may, or may not, be 'developable', depending on a particular set of scenario assumptions.

San Diego Association of Governments (1994). However, it is unique in providing a portable system which can be adapted to any community's GIS data and policy issues.

Using the model

As shown in figure 1, the What if? opening screen contains the standard Windows menu bar, File, and Help options. In addition, it includes three specialized menu options, Suitability, Growth, and Allocation, which correspond to the three procedural steps—determining land-use suitability, projecting land-use demands, and allocating projected demands—incorporated into the model. These options are described briefly below.

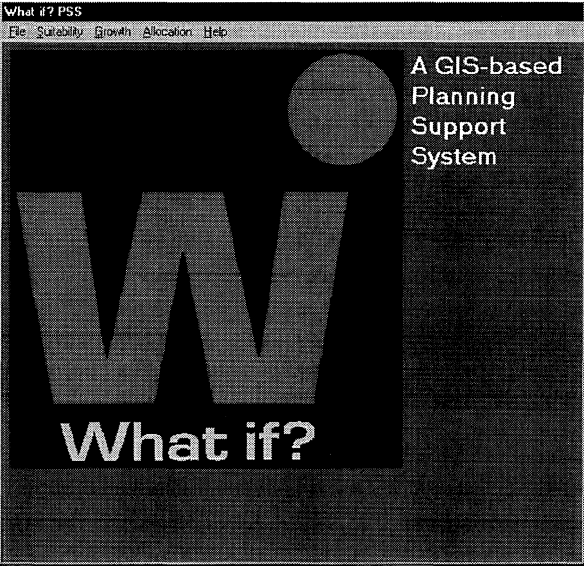


Figure 1. The What if? opening screen.

Determining land-use suitability

The first stage in the What if? analysis process, determining land-use suitability, incorporates standard ‘weighting and rating’ procedures (Hopkins, 1977; McHarg, 1969) that take weeks or months to complete by hand into a quick and easy computer-based process. The suitability analysis process begins by modifying a previously defined suitability scenario or by creating a new one. This is done by selecting the Suitability Define Scenario option from the main What if? menu to access the suitability scenario selection form shown in figure 2. This form can be used to create a new suitability scenario or to review, revise, or delete a previously defined suitability scenario selected from the drop-down list.

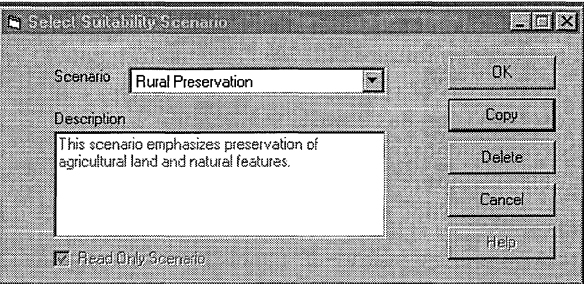


Figure 2. Suitability Scenario selection form.

After selecting a scenario to create, review, or revise, the user is presented with the Suitability Scenario Assumptions form shown in figure 3. As its name suggests, this form is used to specify the assumptions which underlie a particular suitability scenario. The label in the upper left corner of the form identifies the scenario which was selected on the suitability selection form. The drop-down list box at the upper right allows the user to select the land use for which a set of particular suitability assumptions are to be specified. Thus, for example, in figure 3 the user is specifying the assumptions for the 'Low Density Residential' land use for the 'Rural Preservation' suitability scenario.

Figure 3. Suitability Scenario Assumptions form.

As shown in figure 3, the suitability assumptions screen contains four tabbed sheets. These sheets correspond to the four steps of the suitability analysis process: (1) identifying the suitability factors; (2) specifying the suitability factor weights; (3) specifying the suitability factor ratings; and (4) specifying the permissible land-use conversions. These four steps are described briefly below.

Identifying suitability factors The first sheet, labeled 'Factors Considered' and shown in figure 3, contains a series of check boxes which are used to specify the factors which the user feels should be considered in determining the suitability of different locations for a particular land use. Thus, for example, in figure 3 the user has specified that all of the suitability factors should be considered in determining the most suitable locations for low-density residential development. The suitability factors included on the list are determined by the available data for the study area and can include the full range of natural features, for example, slopes, soils, and hazardous areas. Suitability factors must be identified for each land use considered in the study. Different factors may be selected for different land uses; for instance, soils may be assumed to be important for locating retail and office establishments but not for locating single-family developments.

Specifying factor weights The second sheet, labeled 'Factor Weights', contains a series of text boxes that are used to specify the suitability factor weights, that is, the numerical scores indicating the relative importance of different factors for determining the relative

suitability of different locations for a particular land use. Thus, for example, the Slope factor could be given a factor weight of 2 and the Soils factor could be given a factor weight of 1 to indicate that slope is considered to be twice as important as soils in determining a site's suitability for a particular land use. Factor weights are assigned on a five-point scale from 1 (low) to 5 (high).

Specifying factor ratings The third sheet, labeled 'Factor Ratings', contains a series of text boxes that are used to specify ratings for each factor type, that is, for each category within a particular suitability factor (such as for each type of slope). Factor ratings are numerical values which indicate the relative suitability of locations with a particular factor type for a specified land use, for example, the suitability of locations with different slopes for locating low-residential development. The factor types are rated on a six-point scale from 5 (high) to 1 (low) and 0 (excluded). The 'excluded' category identifies areas from which development is to be excluded, regardless of its suitability with respect to other factors. Thus, for example, the user might specify that residential development should be excluded from all locations with slopes which exceed 20%.

Specifying land-use conversions The fourth suitability assumption sheet, labeled 'Conversion', contains a series of check boxes that are used to specify land uses that may be converted from their current use (such as agriculture) to another use (such as low-density residential) during the projection process. If no land uses are identified as conversion candidates, only currently undeveloped land will be available for satisfying the projected land-use demands.

Suitability factors, factor scores, factor weights, and permitted land-use conversions must be specified for all of the land uses considered in the analysis. After this information is provided, the user can click on the Compute button on the suitability assumptions form shown in figure 3. The model then computes the factor scores for each UAZ by multiplying the user-specified factor weights by the corresponding user-defined factor rating and summing these values. The resulting suitability scores indicate the relative suitability of each UAZ for each land use when all of the suitability factors have been considered.

The user can view the suitability analysis results by selecting the Suitability View Scenario option from the main What if? form. The system generates a series of maps showing the relative suitability of different locations for each land use, as shown in figure 4.

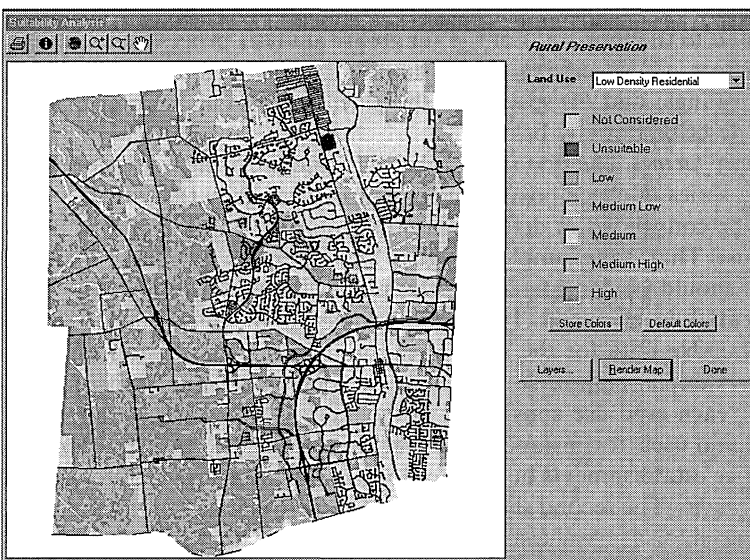


Figure 4. Suitability Analysis map.

The maps can be viewed by selecting the desired land-use category from the drop-down list and clicking on the Render Map button. The maps show each location's suitability for each land use on a scale from 'Not Considered' and 'Unsuitable' to 'High'. The system also generates two reports that can be read on the screen or sent to a printer. The first report identifies the number of acres within each suitability class for all land uses. The second report lists the assumptions which underlie a specified suitability scenario, that is, the factors which were considered in the analysis, the user-specified factor weights and rates, and the permissible land-use conversions.

Projecting land-use demands

What if? considers the demand for land by converting the five main categories of land-use – residential, industrial, commercial, preservation, and locally oriented uses – into the equivalent future land-use demands. The demands are computed for up to five periods, allowing the system to incorporate a staged development process in which future development patterns are based on the previously existing patterns and anticipated infrastructure improvements.

The process of projecting land-use demands begins by selecting the Growth option from the main What if? program menu and by using the growth scenario selection form to create, view, or revise a growth scenario. The form is identical in appearance and function to the Suitability Scenario Selection form described previously. After choosing a growth scenario to view, modify, or create, the user is presented with the Growth Scenario Assumptions form shown in figure 5. The form has five subsidiary sheets which are used to specify the assumptions which define the different land-use demands. The procedures for specifying these assumptions, the growth scenario assumptions forms, and the growth analysis outputs are described briefly below.

Regional Households		
	Year	Number of Households
☐ State Data Center		
☒ Trend	1995	360000
☐ High Probable	2000	365460
☐ Low Probable	2010	371500
☐ Very Low Probable	2020	376540

Share of Regional Households		
	Year	Percentage
☒ Trend		
☐ High Probable	1995	13.3
☐ Low Probable	2000	14.9
	2010	16.5
	2020	18.5

Figure 5. Growth Scenario Assumptions form.

Computing residential demand As shown in figure 5, the Residential sheet contains two tabbed sheets. The first allows the user to select up to five independently prepared projections for the total number of households in the region and the study area's share of the regional households. These two values are multiplied to compute the projected number of households in the study area in each projection period.

The second sheet is used to specify the assumptions about new residential units including: (1) the breakdown by housing type for new residential construction; (2) the housing density for each housing type; (3) the average household size for each housing type; (4) the residential vacancy rates; and (5) the proportion of existing housing units that will be lost to demolition, fire, and so on during each projection period. The model uses the values specified on the two residential sheets to compute the projected demand for residential land in each projection period.

Computing regional business demand The next two sheets, labeled ‘Industrial’ and ‘Commercial’, are used to specify the assumptions which determine the projected demand for regionally oriented land uses. As shown in figure 6, the Industrial sheet allows the user to select between previously developed projections for the total regional employment in each projection year. It also allows the user to specify: (1) the average square footage of floor space per employee; (2) the industrial floor area ratio (FAR); and (3) the vacancy rate for each industrial land use. The Commercial sheet provides similar options for up to five different types of regional commercial development (for example, regional retail, office, and mixed-use areas). These values are used to compute the projected demand for industrial and commercial land uses in each projection year.

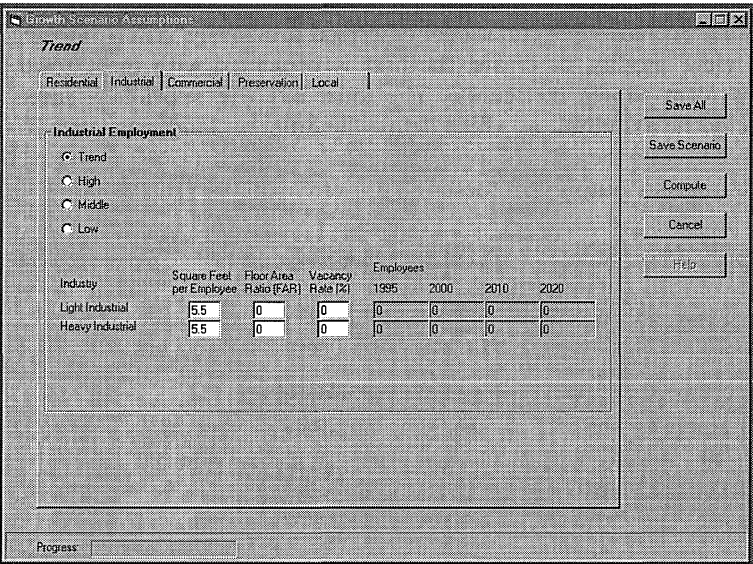


Figure 6. Industrial Growth Scenario Assumptions sheet.

Computing preservation demand The Preservation sheet allows the user to specify the amount of land which should be preserved for environmentally oriented uses such as agriculture, forestry, and open space or environmental protection. The system permits the user to specify the amount of land, either the number of acres or the percentage of the study area, to be allocated for each preservation use in each projection year. These values are treated as land-use demands along with the other land-use demands.

Computing local demands The Local sheet allows the user to specify the quantity of land that will be required per thousand persons to satisfy locally oriented land uses whose location and size are dependent on the local (for example, municipal) population. These uses include local parks and recreational areas, local retail (for example, strip malls), and public and semipublic land uses. The specified land-use standards are applied to the projected population in each political jurisdiction to compute the amount of land which must be devoted to each use.

After providing all the required information on the Growth Scenario Assumptions form, the user can click on the Compute button to calculate the projected demand in each projection year for each land-use type. The system generates a report which identifies the projected demand for each land use in each projection year and a second report listing the assumed values that were used to compute these demands.

Allocating projected land-use demands

What if? projects future land-use patterns by allocating the projected land-use demands derived from a user-selected growth scenario to different locations on the basis of their relative suitability as defined by the assumptions of a user-selected suitability scenario. For instance, the demand for residential land is assigned first to the most suitable sites, then to the second most suitable sites, and so on until all of the projected residential demand in a projection year has been satisfied. If desired, the growth allocation can be controlled by user-selected land-use controls (land-use plans and zoning restrictions) and infrastructure plans. The user is notified if not enough land is available to satisfy the projected demand. If this occurs, the user must modify the suitability, growth, or allocation scenario assumptions (for example, relax the land suitability requirements, allow more land uses to be converted to other uses, or increase development densities) to make them consistent.

The growth allocation process begins by using the allocation scenario selection form to create a new allocation scenario or to revise or delete an existing one. The form is similar in appearance and function to the suitability and growth scenario forms except that it requires the user to identify the suitability and growth scenarios which correspond to a given allocation scenario. The user is then presented with the Allocation Scenario Assumptions form shown in figure 7. The form contains four tabbed sheets which are used to specify the assumptions which underlie a growth allocation scenario, as described briefly below. The first 'Allocation Priority' sheet shown in figure 7, is used to specify the order in which the various projected land-use demands will be satisfied.

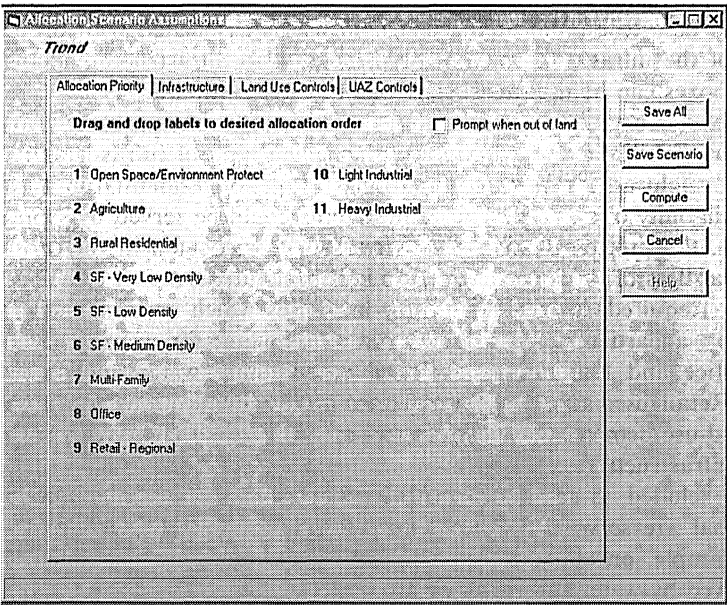


Figure 7. Allocation Scenario Assumptions form.

Infrastructure sheet As shown in figure 8, the 'Infrastructure' sheet contains two tabbed sheets. The 'Infrastructure Plans' sheet is used to select previously defined infrastructure plans which will be used to guide the allocation process. If no infrastructure plans are selected, all of the land uses will be located in their most suitable locations, regardless of the availability of sewer and water service or their proximity to major roads or intersections.

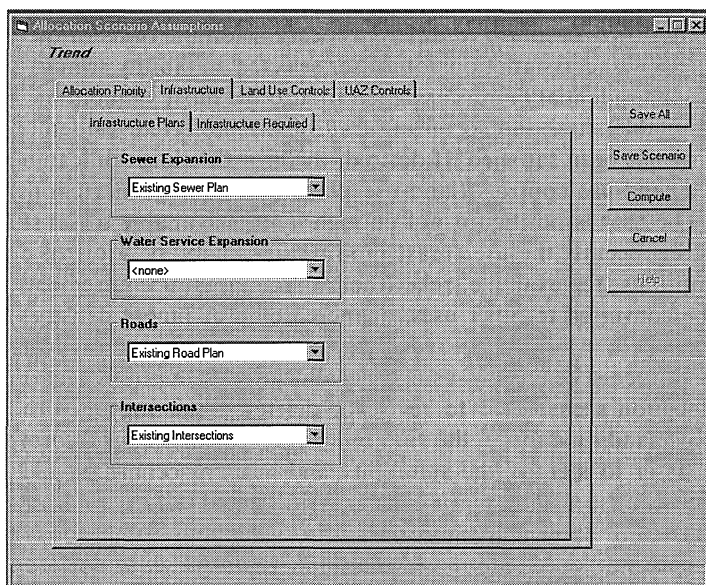


Figure 8. Infrastructure Plans sheet.

The 'Infrastructure Required' sheet is used to specify the assumed relationship between infrastructure provision (or proximity) and land-use development patterns. The sheet lists the land uses to be allocated; each land use is associated with drop-down boxes for each of the infrastructure controls listed on the Infrastructure Plans sheet. The drop-down boxes can be used to specify one of three options.

The first option, Not Affected, is used for land uses whose location is assumed not to be affected by the availability of (or proximity to) a given type of infrastructure. Thus, for instance, one might assume that the location of large-lot residential units is not affected by the availability of sewers because these units could be served by septic systems. In these cases, the projected land-use demands will be located in the most desirable locations, regardless of the presence of this infrastructure.

The second option, Required, is used to identify land uses which are assumed to require—or be near to—a particular type of infrastructure. Thus, for example, one could assume that Office and Neighborhood Retail uses require water and sewer service and Regional Retail uses must be located near major intersections. In these cases, the specified land-use demand will be allocated only to areas that have (or are near to) the specified infrastructure.

The third option, Excluded, is used to identify land uses that are assumed to be excluded from areas that are served by (or are near to) a particular type of infrastructure. Thus, for example, one could assume that agriculture would be excluded from areas that are served by water and sewer or are near major intersections because these locations will be too expensive to be devoted to agricultural uses. In these cases, future land-use demands will be excluded from areas that are served by, or are near to,

a particular type of infrastructure. The final sheet, labelled 'UAZ Controls', contains a series of text boxes that are used to specify the minimum-sized UAZs into which different land-use demands can be allocated, for example, that industrial development cannot take place in UAZs smaller than, say, five acres in size.

Land Use Controls sheet This sheet is similar to the Infrastructure Plans sheet shown in figure 8 and allows the user to select between previously defined land-use plans and zoning ordinances which will control the location of future growth. If a land-use control ordinance or plan is selected, the system allows projected land-use demands to be allocated only to areas in which the land use is permitted or planned. If no land-use control plan is selected, land-use demands will be allocated to their most suitable location, without any restrictions.

When the user clicks on the Compute button the system projects future land-use patterns by allocating the projected land-use demands to the most suitable sites, subject to any land-use controls or infrastructure plans selected by the user. The resulting land-use projections for each projection year can then be viewed in map or report form.

For example, figure 9 shows the projected land-use patterns for a 'Compact Allocation' scenario whose user-specified suitability ratings and allocation controls have concentrated future residential and commercial development largely in the eastern part of the study area. Figure 10 (over) shows the projected land-use patterns for a 'Dispersed Allocation' scenario which incorporates the same land-use demand assumptions and different infrastructure and land-use control assumptions which allow development to be spread throughout the study area. These two examples illustrate the power of tools such as What if? for graphically portraying the implications of alternative policy choices. The system also generates reports which record the projected land-use quantities for the study area and each political subdivision in each projection year and the assumptions which underlie a scenario.

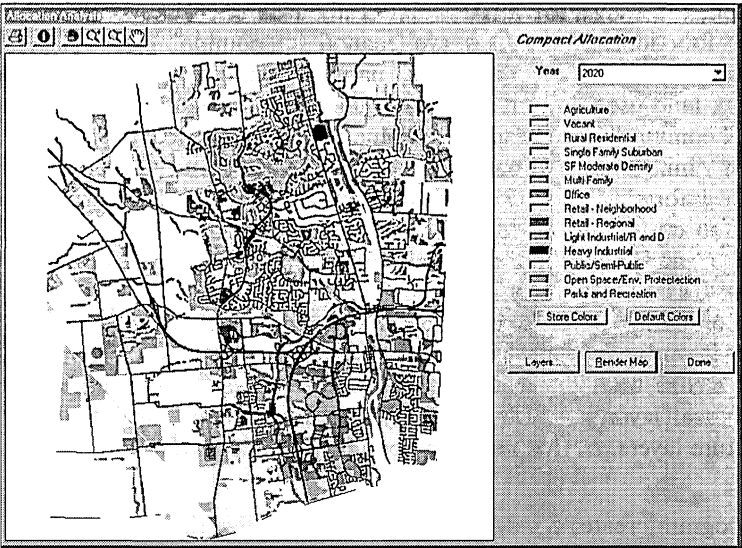


Figure 9. Allocation map for 'Compact Allocation' scenario.

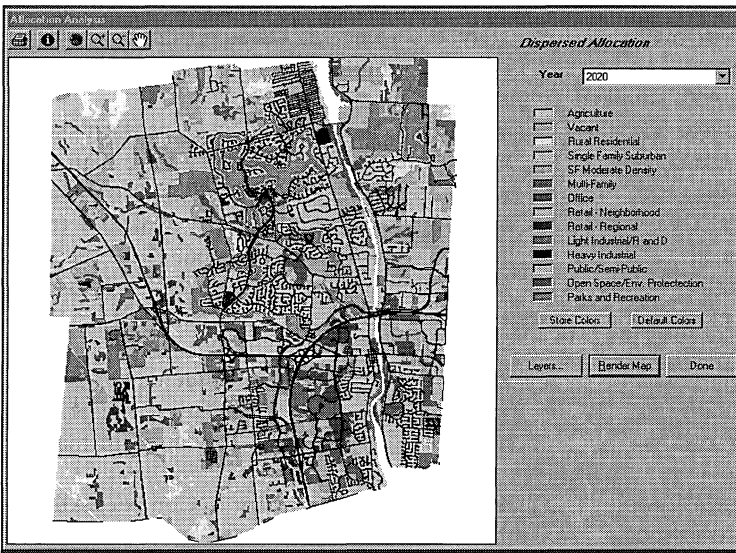


Figure 10. Allocation map for 'Dispersed Allocation' scenario.

Installing What if?

Although it incorporates many of the concepts of other GIS-based models such as the first California Urban Futures Model (Landis, 1994; 1995) What if? is unique in its ability to be adapted to the GIS data sets and policy concerns for one or more study areas. These capabilities are provided by its GIS-created UAZ files, Setup, and Manager programs which are described briefly below.

Creating UAZ files

Before What if? is used, the UAZs for a particular study area must be created with standard GIS overlay functions. The UAZs are created by combining GIS layers containing information on, among other things, natural conditions, existing and proposed infrastructure, land-use controls, and the boundaries for subdivisions of the study area. GIS UNION commands are used to combine these layers into a single layer made up of UAZs containing information from each of the constituent layers. That is, each UAZ contains information on the natural features, available infrastructure, planned land use, and so on for all points lying inside of it. The UAZs are stored in ESRI's 'shape file' format and incorporated directly into the What if? program.

The only GIS layer that is absolutely essential for using What if? is the existing land uses. A variety of additional layers can be added, depending on the available GIS data, the user's analysis and policy needs, and the requirements of any secondary applications that will utilize the What if? outputs. In most cases it is desirable to have at least three natural layers for the suitability analysis, two or three land-use control and infrastructure layers, and some basic boundary display layers.

Setup program

The What if? Setup program creates the system files that customize the software for a particular collection of GIS (and non-GIS) data. As shown in figure 11, the main Setup form contains a series of options that are used to define the land-use categories, suitability factors, and factor types that will be used in the analysis and the labels and the other information that appears on the suitability, growth, and allocation scenario assumption forms.

The user can enter the required information by clicking on one of the buttons displayed along the left-hand side of the main Setup form. For example, clicking on the Define Factors button displays the form shown in figure 12. This form specifies the labels for the suitability factors that will be used in a particular analysis, as determined by the GIS data that are available for the study area. Similar forms define the land uses in the study and the labels that appear on the suitability, growth, and allocation forms. After entering all of the required data, the user can select the 'Create System Files' option at the bottom of the form to create a set of customized scenario assumption forms containing the labels defined by the user. These forms can then be used to create the suitability, growth, and allocation scenarios as described previously.

What if? Setup

File

WHCPSS

- ☒ **General Information**
 - ☒ Define Suitability Land Use Categories
 - ☒ Define Existing Land Use Categories
 - ☒ Define Allocation Land Use Categories
 - ☒ Define Planned Land Use Categories
- ☒ Map ELUs to ALUs
- ☒ Map ALUs to SLUs
- ☒ Map ALUs to ELUs
- ☒ Map ALUs to PLUs
- ☒ Define Projection Periods
- ☒ **Suitability Information**
 - ☒ Define Factors
 - ☒ Define Factor Types
- ☒ **Growth Information**
 - ☒ Define Sub-Areas
 - ☒ Define Residential Information
 - ☒ Define Commercial Information
 - ☒ Define Industrial Information
- ☒ **Allocation Information**
 - ☒ Define Infrastructure Types
 - ☒ Define Infrastructure Controls
 - ☒ Define Land Use Controls
- ☒ **Completed**
 - ☒ Create System Files

Figure 11. What if? Setup form.

Manager program

The What if? Manager program can be used to revise or delete a set of scenario assumption forms once they have been created. It allows the user to modify the labels that appear on the scenario assumptions forms (to, for instance, correct any spelling errors), modify the names used to identify different land-use categories, or change system-defined variables such as the household and employment projections. These corrections can be made after the forms have been used to create a set of scenario assumptions.

Evaluation

What if? has many obvious strengths, many of which are equally obvious weaknesses. Its biggest advantage is that it is a fully operational model which can be adapted to the particular data sets and policy concerns in any area which satisfies the rather modest

Define Factors

Column	Name
Factor 1	Slopes
Factor 2	Soils
Factor 3	100-yr Flood plain
Factor 4	Flooding Potential
Factor 5	Endangered Species
Factor 6	Stream Buffer
Factor 7	
Factor 8	
Factor 9	
Factor 10	

Cancel OK

Figure 12. Define Factors form.

data requirements of the model. These requirements include: (1) a required GIS layer containing current land uses; (2) optional layers describing natural features, administrative boundaries, and land-use and infrastructure plans; and (3) projected, or assumed, values for the region's projected population and employment, household and employment densities, and the like. These data requirements are particularly modest in comparison with those of other operational urban models such as DRAM/EMPAL (Putman, 1983; 1992), MEPLAN (Echenique et al, 1990), the CUF models (Landis, 1994; Landis and Zhang, 1998), and UrbanSim (Waddell, 1998).

A second obvious advantage is the simple and intuitive modeling structure of What if? The general concepts of balancing the supply of, and demand for, land by determining the relative suitability of different locations, projecting the demand for land, and allocating the projected demands to the most suitable locations is easily understood by elected officials and the public. The detailed computations required to carry out these three stages are of course too complex for most users to understand fully. However, the underlying structure of the model is much easier to understand than the other, much more complex urban models listed above.

What if? shares many of the advantages (and disadvantages) of the CUF model on which it is based (Landis, 1994, pages 400–402, 418–419; 1995, page 444). Like the CUF model—and unlike other operational urban models—What if? is GIS based. This allows the model to take advantage of the diversity of spatially referenced data which are available in a particular study area and grow as more data become available. The use of spatially disaggregated GIS data also allows the model to allocate land-use demands to particular sites and not to spatial aggregates such as census tracts or traffic analysis zones. Again, like the CUF model, What if? is policy oriented and simulates alternative development patterns as determined by local land development policies. Unlike traditional urban models, the critical variables which determine different model outcomes are local land development policies, not abstract 'secondhand' effects of these policies such as changes in travel times or accessibility.

The simple structure and minimal data requirements of What if? are associated directly with its most glaring weakness, its lack of a firm theoretical basis. The model does not include measures of spatial interaction, that is, accessibility to employment, shopping, and recreational opportunities which are widely recognized to be an important—if not the most important—determinant of long-term urban growth patterns and the key component of most urban models. Unlike the acknowledged 'state of the art' in urban models (Wegener, 1995, page 22) it does not rely on random utility or discrete choice theory to explain and predict the behavior of urban actors such as households, investors, and firms. It also does not represent the interlinked markets for land, housing, nonresidential uses, labor, and infrastructure, or provide any procedures for 'market clearing' and price adjustment in the face of changes in demand and/or supply. In addition What if? also does not explicitly model the behavior of actors such as households, businesses, and developers in the same way as 'object-oriented' models such as UrbanSim (Waddell, 1998).

Instead of incorporating any of these theoretically appealing, but complex, concepts, What if? has been designed to provide a plausible, intuitive, and adaptable fully operational model of the urban development process. The behaviors of urban markets and actors are represented implicitly in the assumption that land uses will be allocated to their most suitable location, as determined by a set of user-specified suitability assumptions. Similarly, the impact of alternative growth management policies is incorporated into the model as an assumption that development will occur only in places where it is permitted under a zoning ordinance or anticipated in a land-use plan. Neither assumption begins to capture the complexities of actual urban development

processes. However, they do provide a readily understandable and plausible basis for considering the implications of alternative public policies. The overarching goal, after all, is not projecting the future exactly but providing a meaningful foundation for community dialogue and debate.

The ideal PSS will provide an "intelligent digital toolbox" that helps users select the most appropriate software tools from a range of alternative analysis and forecasting models, applies these tools to data which have been accessed locally or via the Internet, and allows users to view quickly the implications of alternative policy choices in intuitive graphic, map, and interactive video and/or sound displays (Klosterman, 1997, page 52). What if? is only one of a number of systems that are being developed to help achieve this ideal (Klosterman, 1999). However, it is unique in providing a readily available package that is easy to use and understand, fully customizable to a community's existing GIS data set, and incorporates the suitability, growth, and allocation portions of the planning process. As a result, it may provide a useful component of a more comprehensive PSS which considers a wide range of land-use-related issues such as the fiscal, transportation, and environmental impacts of alternative growth scenarios.

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